# OVERVIEW ANATOMY OF URINARY SYSTEM

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**Anatomy, DMC**

# LEARNING OBJECTIVES

**At the end of the class the student should be able to**

- Discuss the components of the urinary system
- Discuss the gross anatomy of the kidney
- Know the histology of the kidney
- Discuss the development of kidney
- Discuss the blood supply venous drainage and nerve supply of kidney

# Components of Urinary System

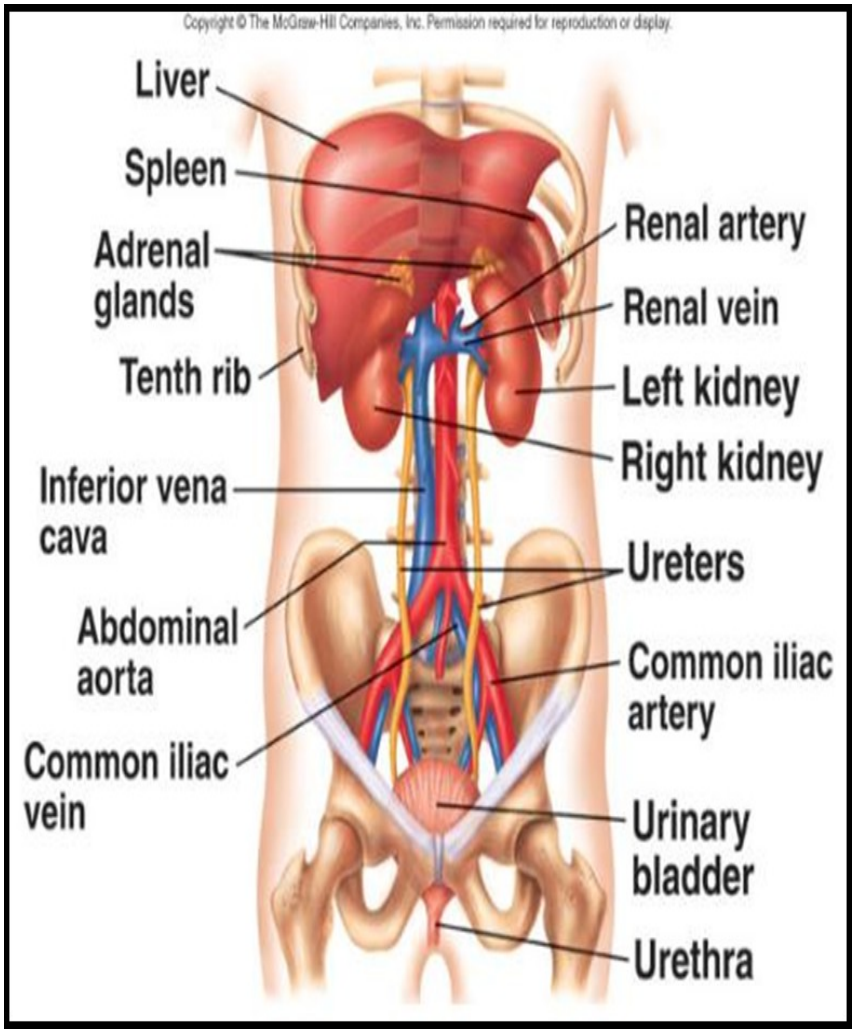
The Urinary System comprise

**Kidneys** where urine manufactures and secreted

**Ureters** acts the channel to transport the urine

**Urinary Bladder** receives the urine through the ureter and retained

**Urethra** tube like structure, longer in male used to excrete the urine.



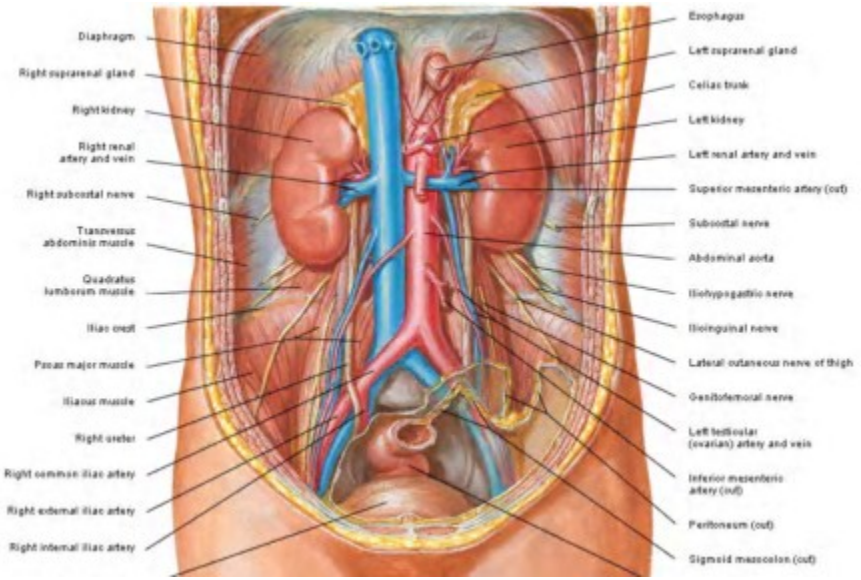
# Gross Anatomy of Kidneys

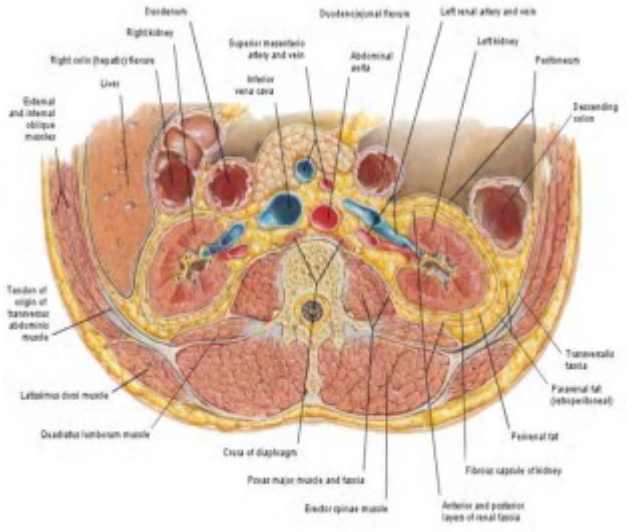
Kidneys are paired organs

Situated on posterior abdominal wall

One on each side of the vertebral column behind peritoneum.

Long axis of kidney is directed downwards and laterally so upper poles are nearer to the median plane



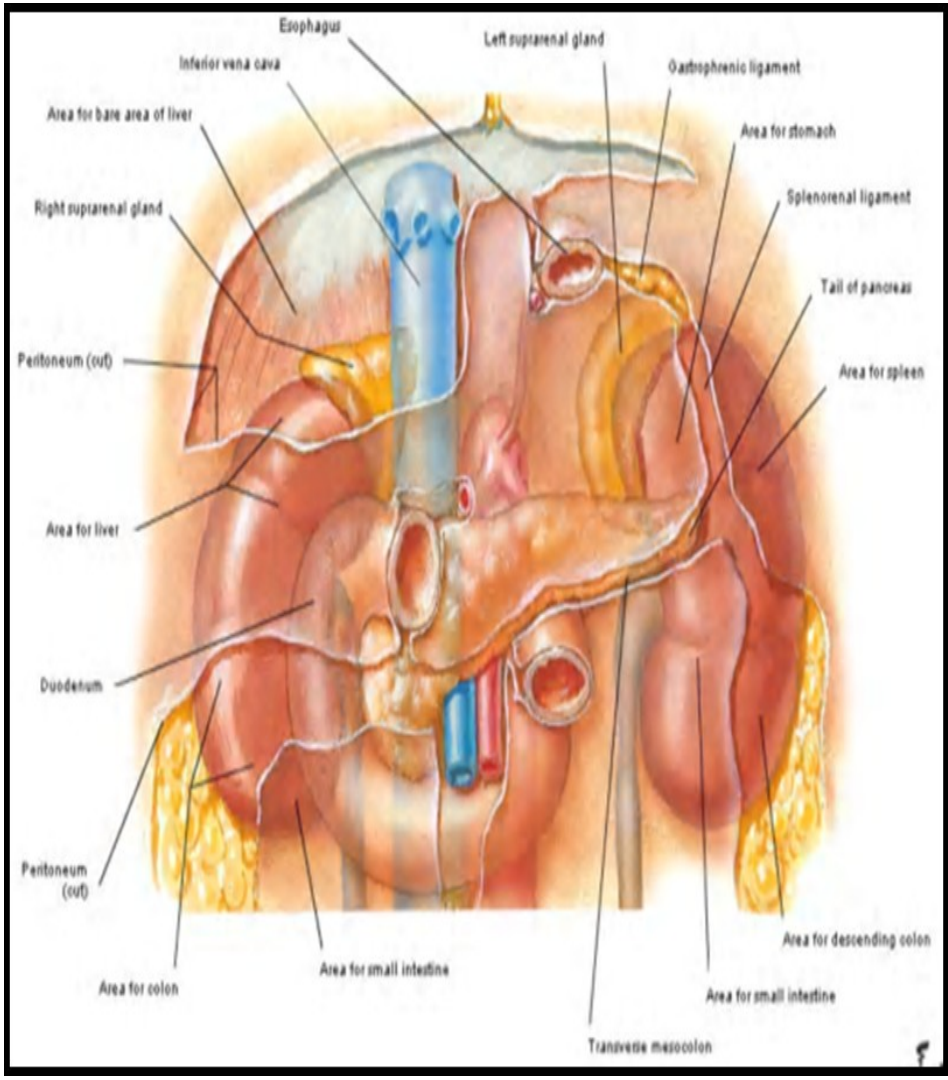
# Dimension

Each kidney is **11cm** in **length** **6 cm** in breadth and **3cm** in anterioposterior diameter

Weighs about **150 gm** in male and **135 gm** in female.

The right kidney is lower as compared to left because of liver.

The hilum the of left kidney is just above and right kidney is just below the trans-pyloric plane.



# Appearance

Kidney has two poles

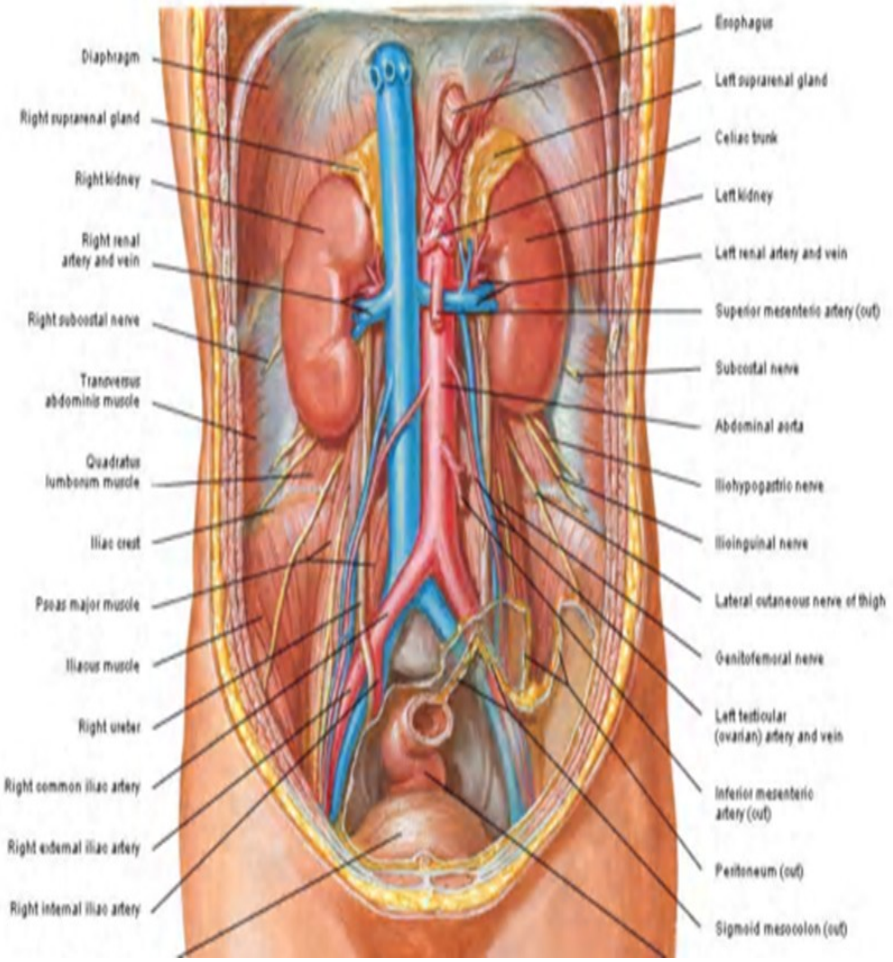
A **upper pole** which is broad and is in close contact with the corresponding suprarenal gland

A **lower pole** which is pointed

Kidney has two surfaces

The **anterior surface** is irregular

The **posterior surface** is flat



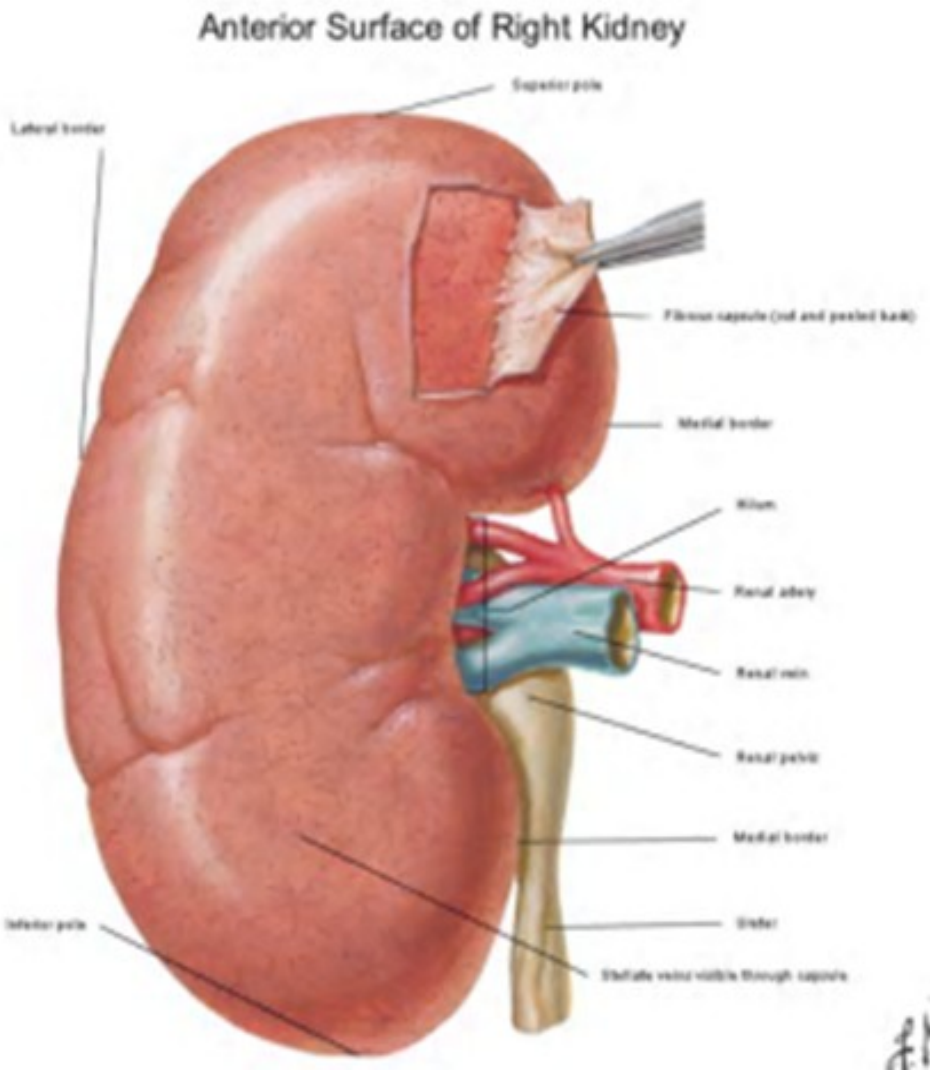
# Appearance

Two Borders

**Lateral border** which is convex

**Medial border** which is concave having in its middle part a depression the **hilus** or **hilum**

Arrangement of structures at hilum from anterior to posterior is Renal vein , Renal artery, Renal pelvis



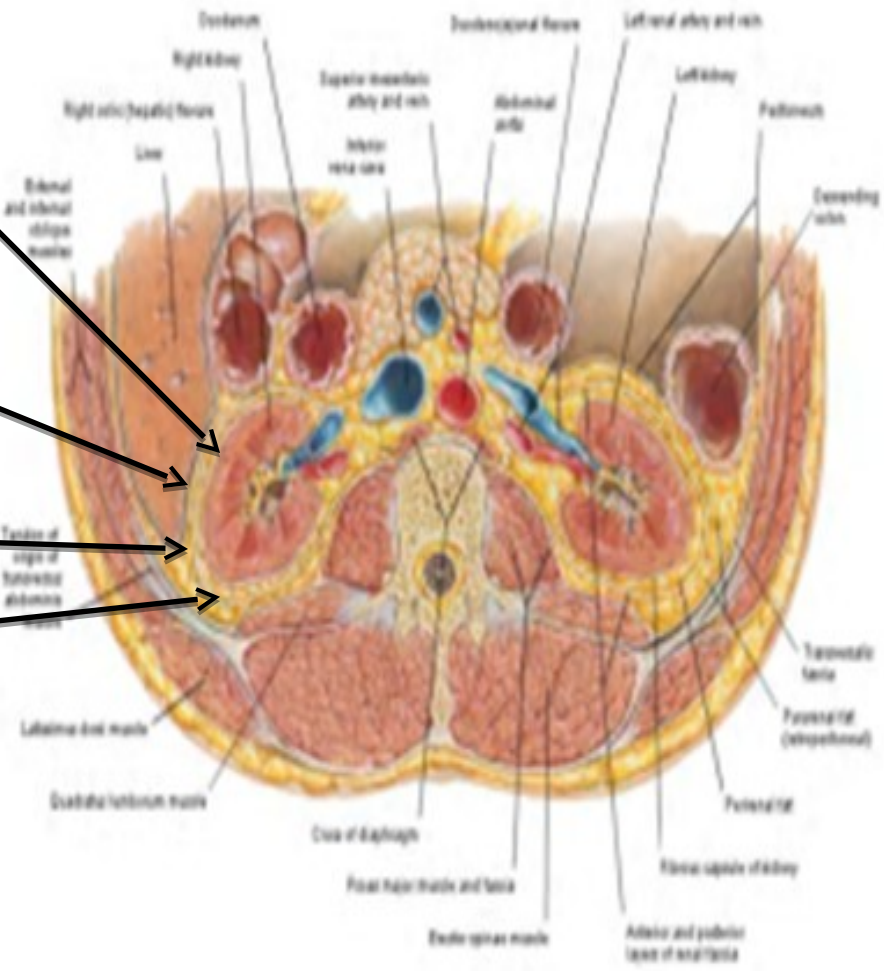
# Coverings of Kidney

**Fibrous capsule:** is thin membrane which invests kidneys and lines the renal sinus

**Perinephric fat:** Is a layer of adipose tissue out side fibrous capsule filling up the extra space in the renal sinus

**Renal fascia:** Is a fibroalveolar sheath outer to perinephric fat.

**Paranephric body fat:** Is the fat out side the renal fascia. More abundant posteriorly and lower pole of kidney. It fills up the paravertebral gutter and forms cushion for kidney



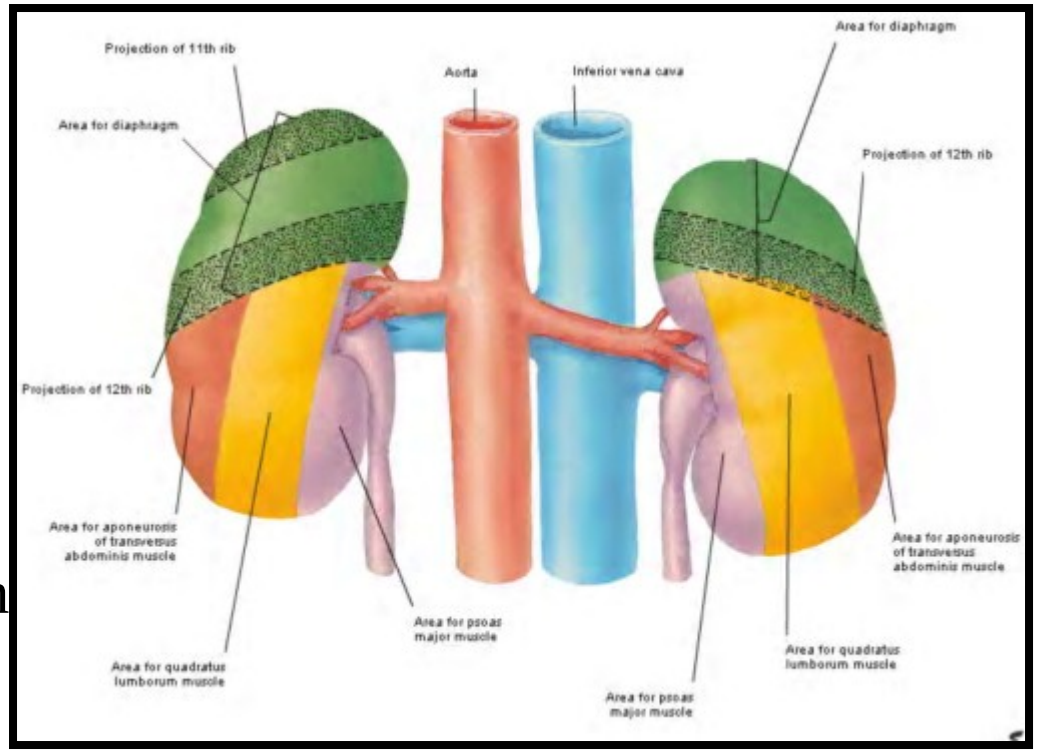
# Relations of Right Kidney

**Anterior relations:**

- Right supra renal gland
- Liver
- Second part of duodenum
- Hepatic flexure of colon
- Small intestine

**Posterior relations:**

- Diaphragm
- Medial and lateral arcuate ligaments
- Psoas major and quadratus lumborum and transverse abdomin muscles
- Subcostal vessels
- Subcostal, iliohypogastric and ilioinguinal nerves
- Twelfth rib



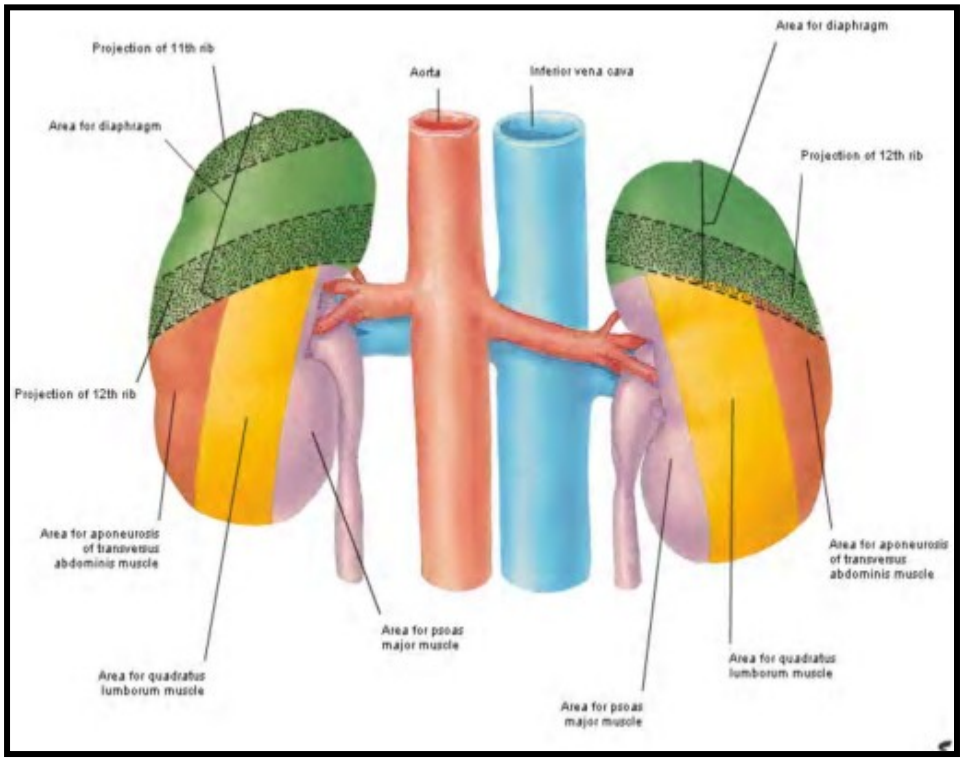
# Relations of Left Kidney

**Anterior relations :**

- Left supra renal
- Spleen
- Stomach
- Pancreas
- Splenic vessels
- Splenic flexure of colon and descending colon
- Jejunum

**Posterior relations:**

- Diaphragm
- Medial and lateral arcuate ligaments
- Psoas major and quadratus lumborum and transverse abdominal muscles
- Subcostal vessels
- Subcostal, iliohypogastric and ilioinguinal nerves
- Eleventh rib
- Twelth rib



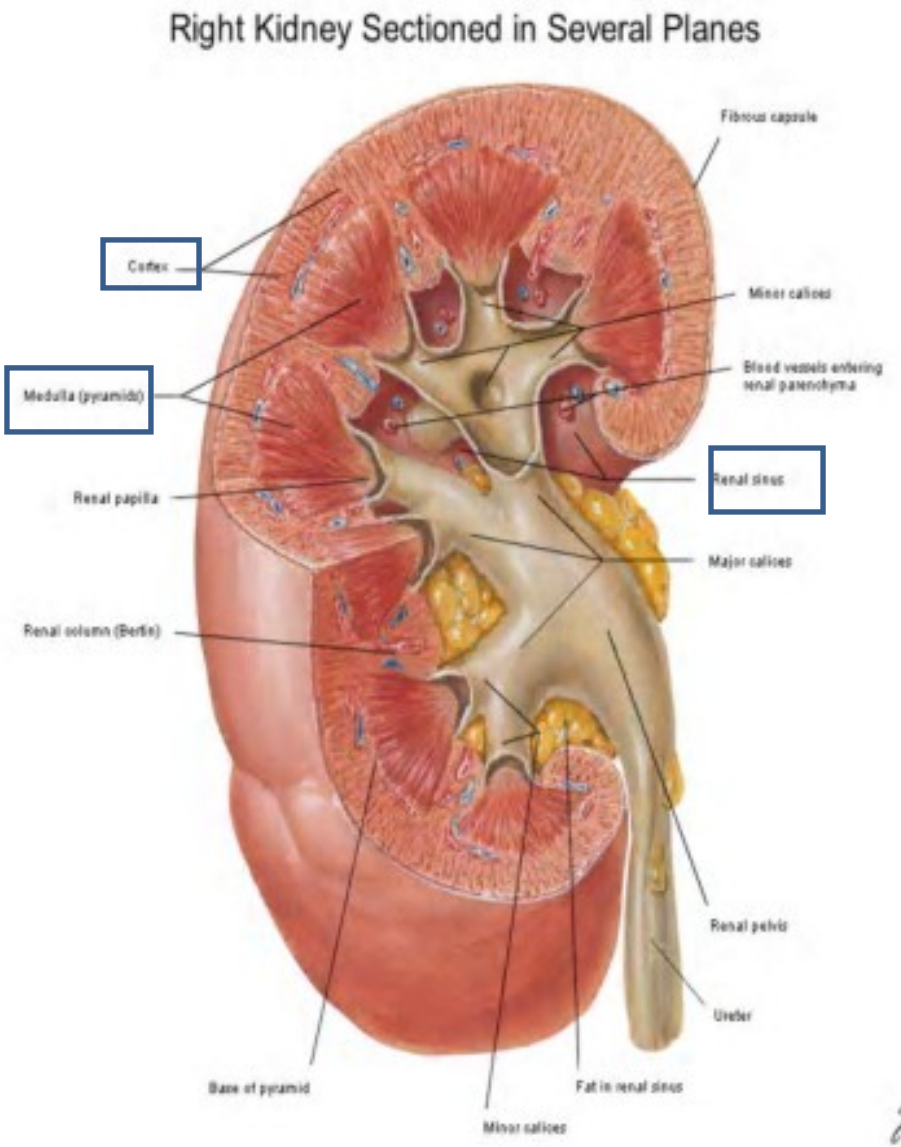
# Coronal Section of Kidney

The coronal section of kidney shows the:

- Outer reddish brown cortex
- Inner pale medulla
- A space renal sinus.

**Renal cortex** is divisible in two parts

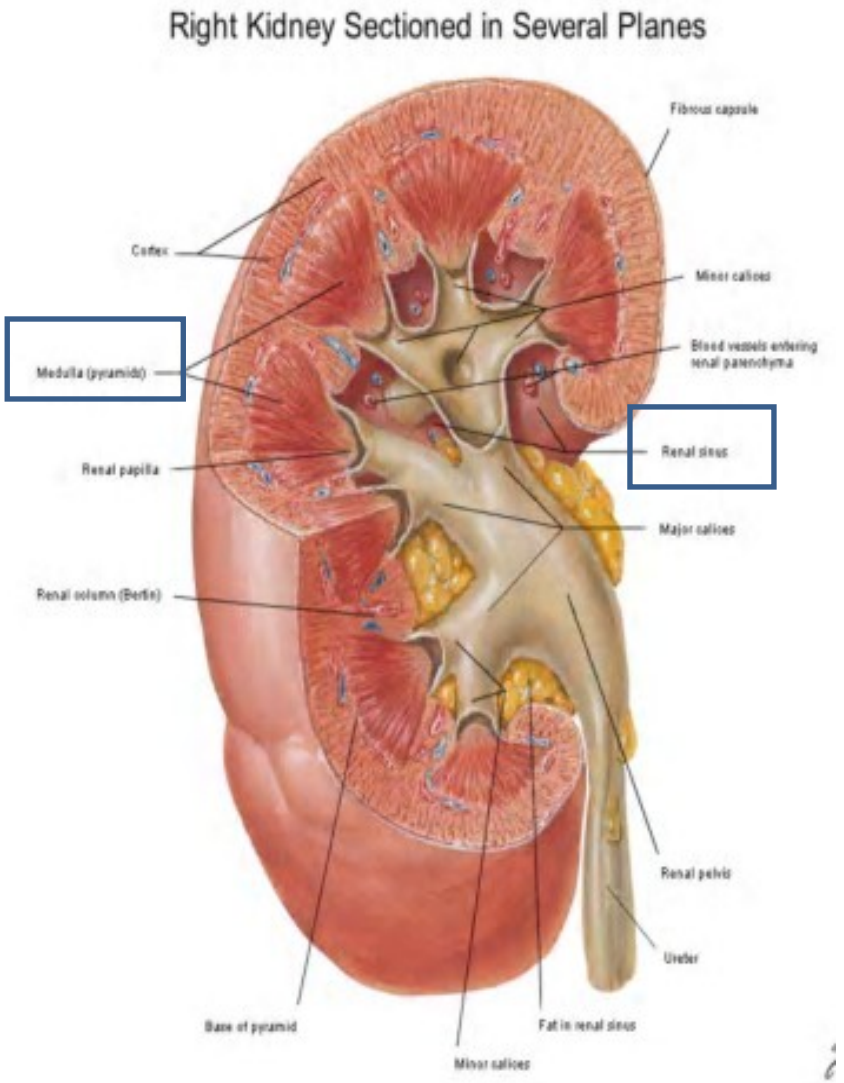
- Cortical arches or cortical lobules.
- Renal columns which dip in between pyramids. Each pyramid along with overlying cortical arch form a lobule



# Coronal Section of Kidney

**Medulla** is made up of about 10 conical masses called renal pyramids. Apices of pyramids form renal papilla which indents minor calices.

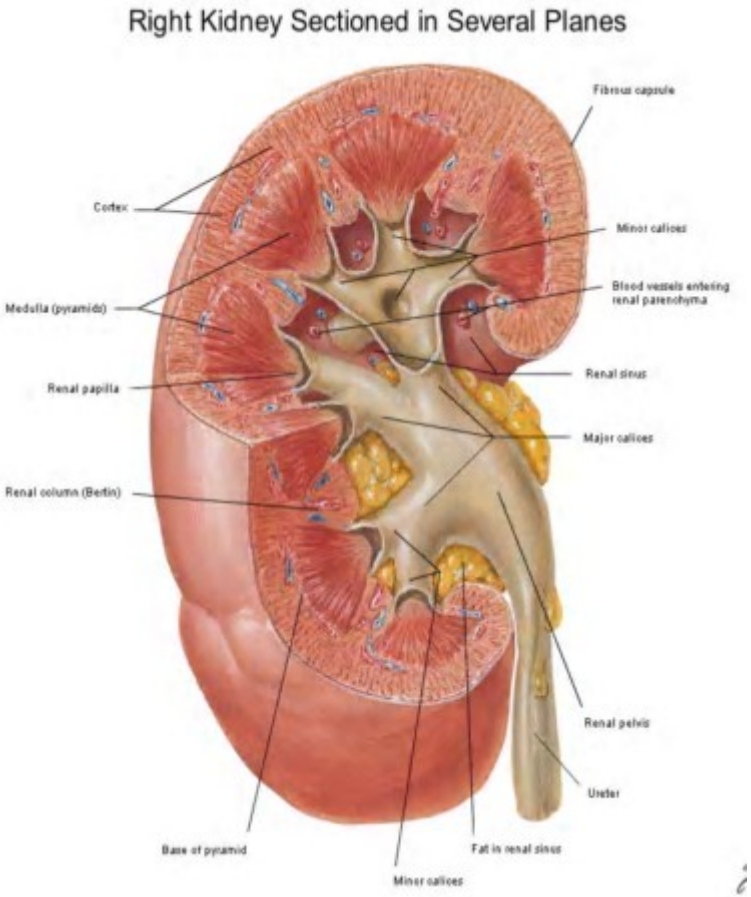
**Renal sinus** is a space that extends into the kidney from hilus contains renal artery, tributaries of renal vein and renal pelvis (pelvis divides into two to three major calices and these in turn divide into 7 to 13 minor calices and each minor calyx ends into an expansion indented by one to three renal papillae).



# Histology of Kidney

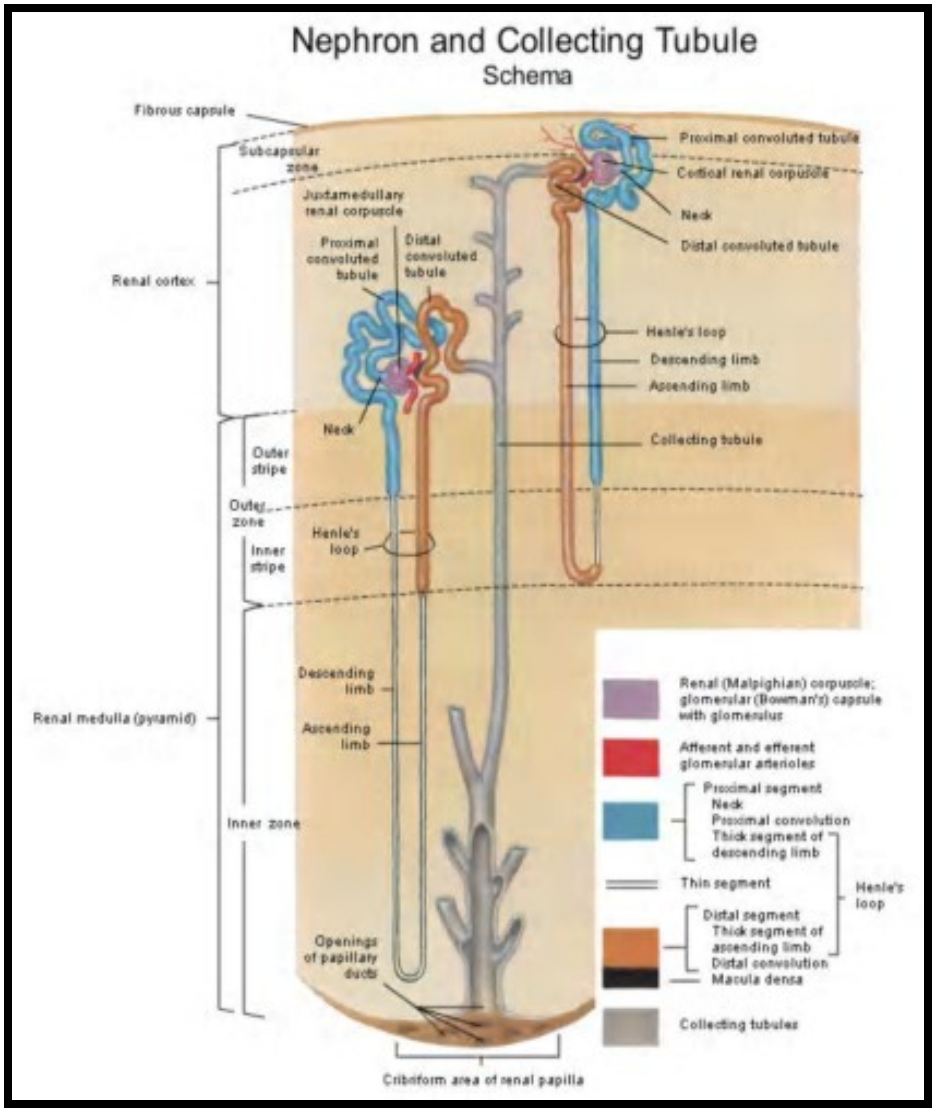
Kidney is compound tubular gland having outer cortex and inner medulla.

- **Cortex:** Is granular because it contains renal corpuscles (i.e. bowman's capsule, and glomerulus). and convoluted part of renal tubules. Renal cortical substance extending in to medulla is called renal columns.
- **Medulla:** Has striated appearance because it lodges parallel arranged blood vessels and straight parts of rena tubule. Medulla extending into cortex is called medullary rays
- **Pyramid:** Medulla has 8-10 conical masses whose base is at the cortex and apex (papilla) projects in to minor calyx.
- **Lobes of kidney:** Each pyramid with its associated overlying cortex is regarded as a lobe of kidney.



# Unit of Kidney

- The structural and functional unit of kidney is nephrone
- Each kidney contains approximately one million nephrons
- Each nephron begins as double walled cup shaped dilatation the bowman’s capsule having a parietal and visceral layer and a space called urinary space



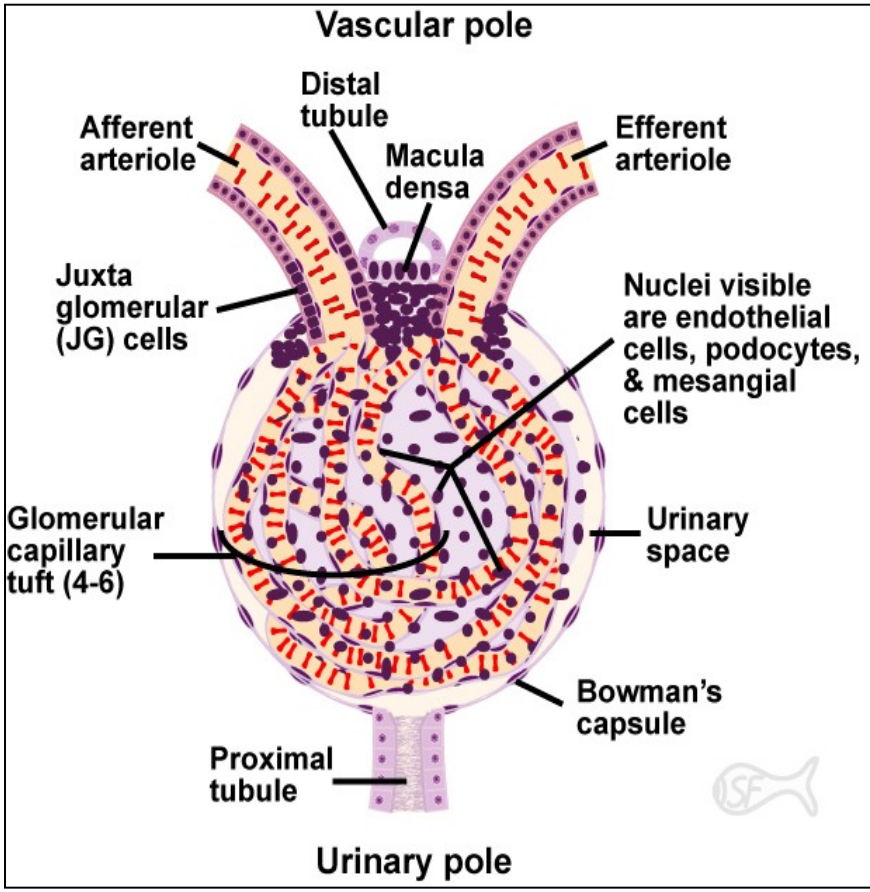
# Bowman Capsule

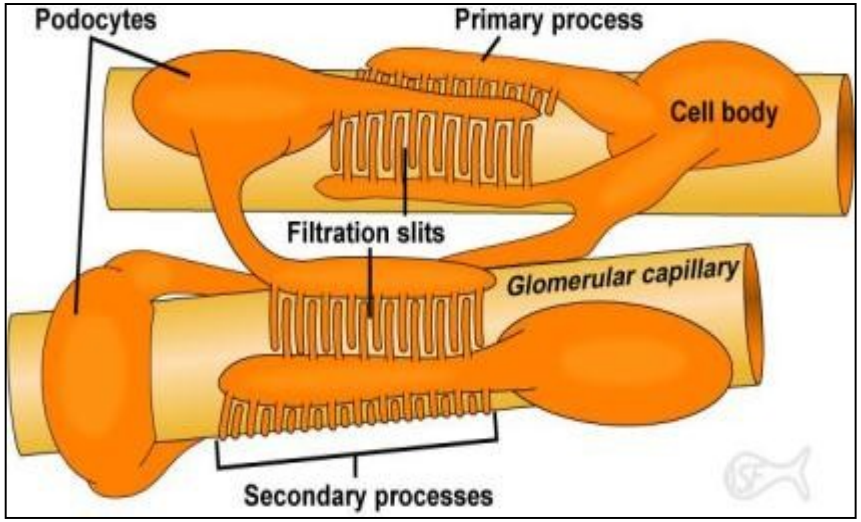
**Parietal layer** of bowman capsule is lined by simple squamous epithelium which continues with low columnar epithelium of proximal convoluted tubule.

**Visceral layer** of bowman's capsule: Covers glomerulus and Invests it closely and is composed of star shaped epithelial cells (single layer) called podocytes which share basal lamina with endothelial cells of capillaries.

From cell bodies arise several primary processes resembling tentacles of octopus extending towards glomerular capillary loops and give rise to numerous secondary branches called foot processes or pedicles.

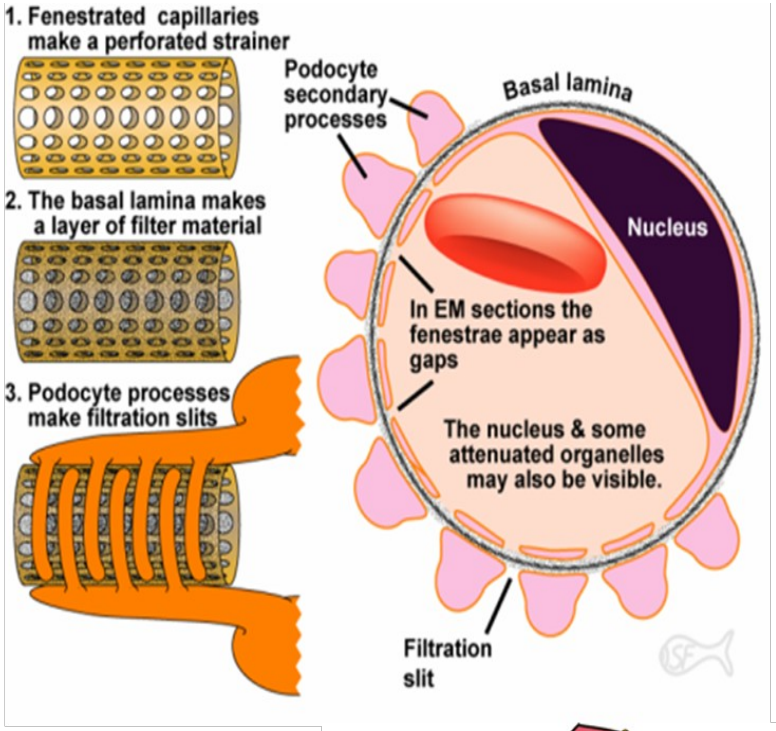
Pedicles make a direct contact with the capsular space of common basal lamina.

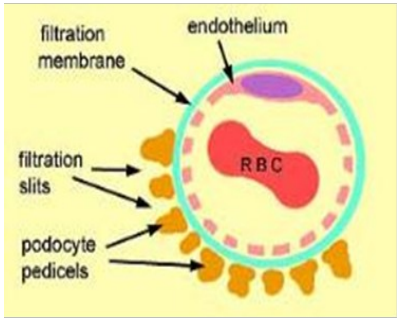


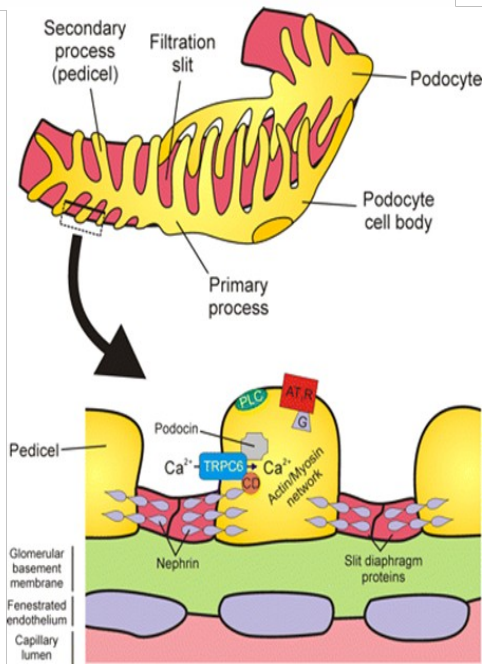



# Filtration Slit

- The adjacent pedicles are separated by narrow spaces called **filtration slits** (about 30 nm).
- Filtration silts are spanned by very thin membranes or silt diaphragm (5 nm thick).
- **Silt diaphragms** are porous (average pore diameter 6 nm) so it prevents passage of larger molecules.
- **Basal lamina of visceral layer:**
- Lies between podocytes and endothelial cells of glomerular capillaries.
- It is common for podocytes and glomerular capillaries. In children 150nm and in adults 300 nm, continuous, uninterrupted layer having no gaps or pores.



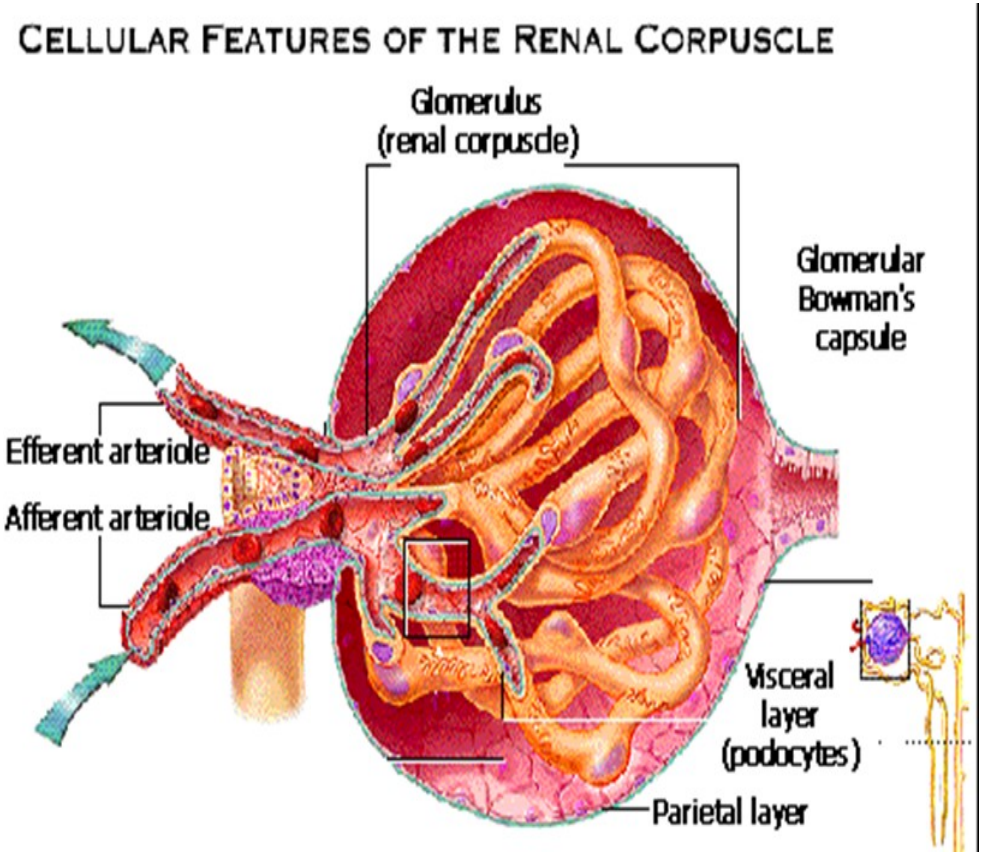



# Glomerulus

Consists of **capillary loops** connecting afferent and efferent arteriole.

Two arterioles lie close together at **vascular pole** of the renal corpuscle.

Afferent arteriole divides 4-5 main branches each of which divide into number of capillaries which join together to form efferent arteriole. Efferent arteriole contains greater amount of smooth muscles in its walls, so its lumen is narrower than afferent arteriole



# Proximal convoluted tubule

Longest and most tortuous (**14 mm** long). Starts at urinary pole and ends by becoming continuous with loops of Henle.

**Under light microscope:**

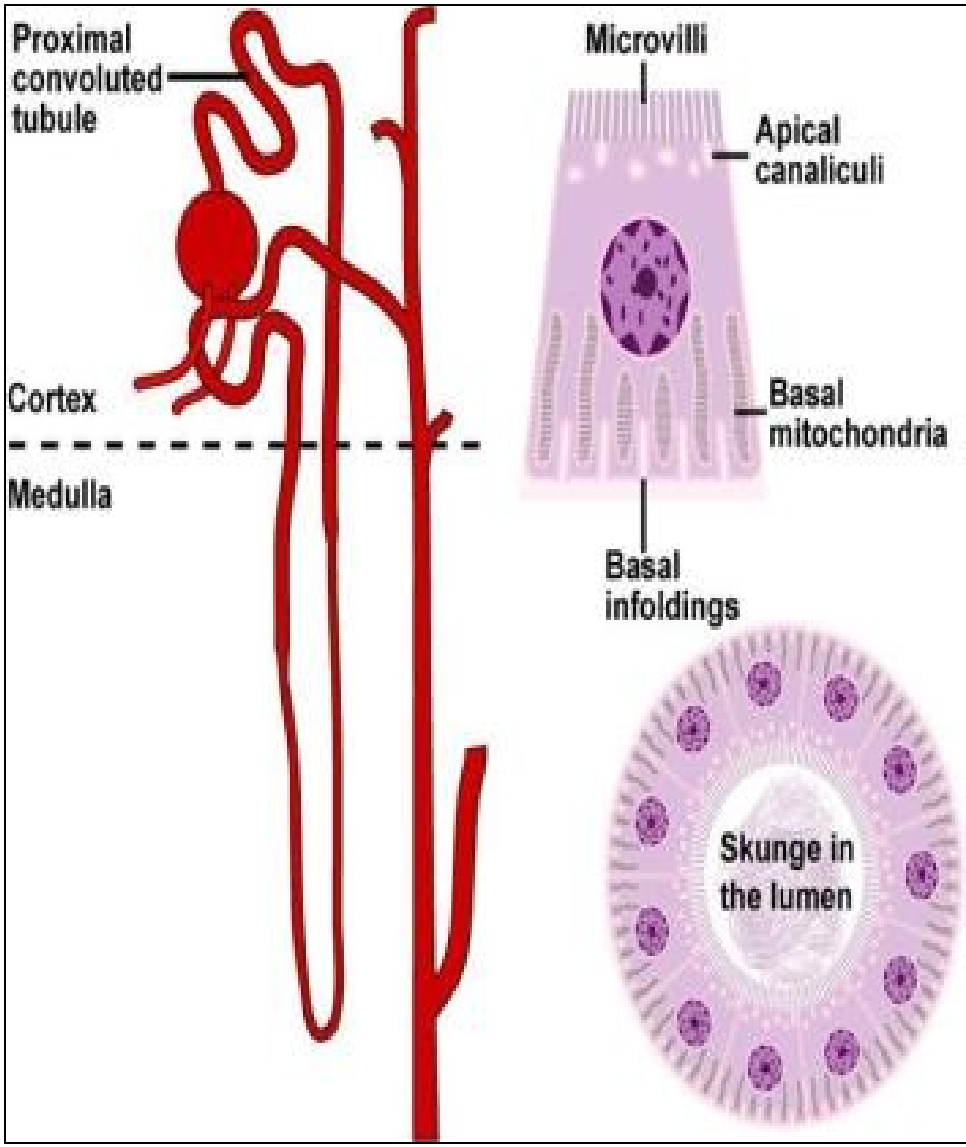
Lined by **low columnar epithelium with c**entrally located spherical nucleus.

Free surface of cells show brush border.

**Under electron microscope:**

**Brush border** is composed of long closely packed microvilli, covered by thick layer of glycocalyx. **brush border** provide 20 fold amplication of surface area of cell

Abundant mitochondria provide energy, present at basal margin producing Basal infolding



# Loop of Henle

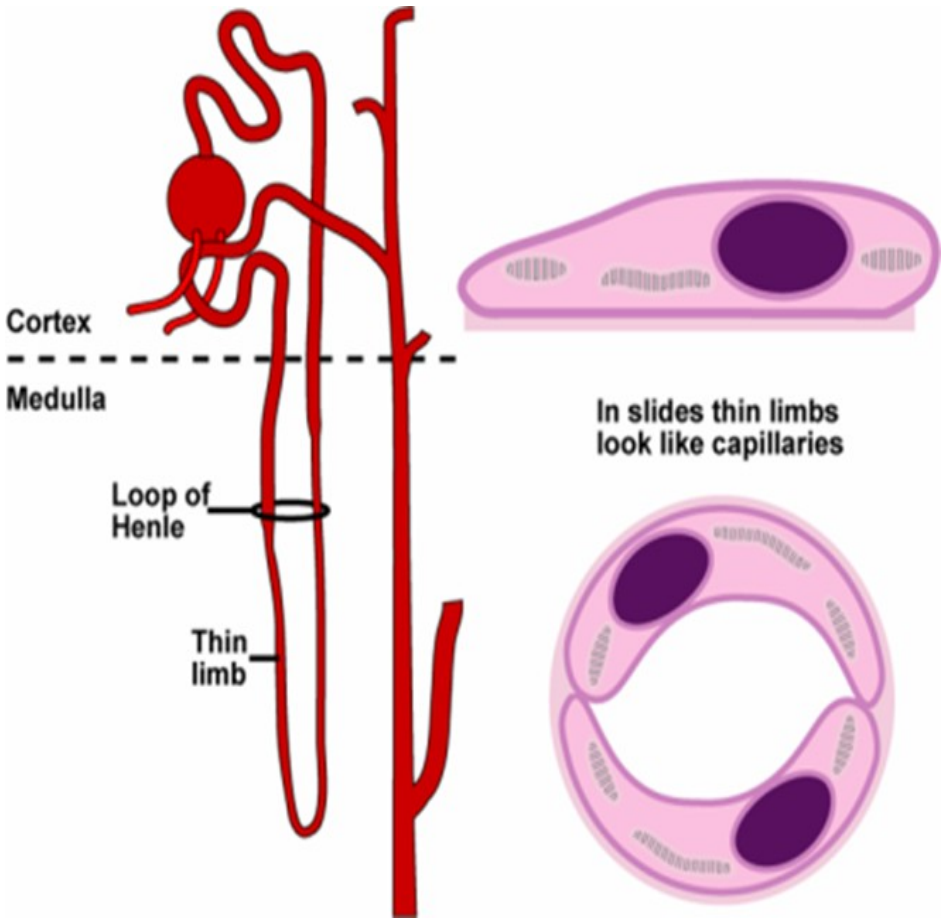
Consists of **proximal straight tubule**, **thin segment** and **distal straight tubule**.

In **cortical nephrons** has shorter loop of Henle which extend only slightly in to the medulla or are confined to medullary rays.

In **justamedullary nephrons** (15%) are located near cortico-medullary junction have very long loop of Henle which extend deep into the medulla.

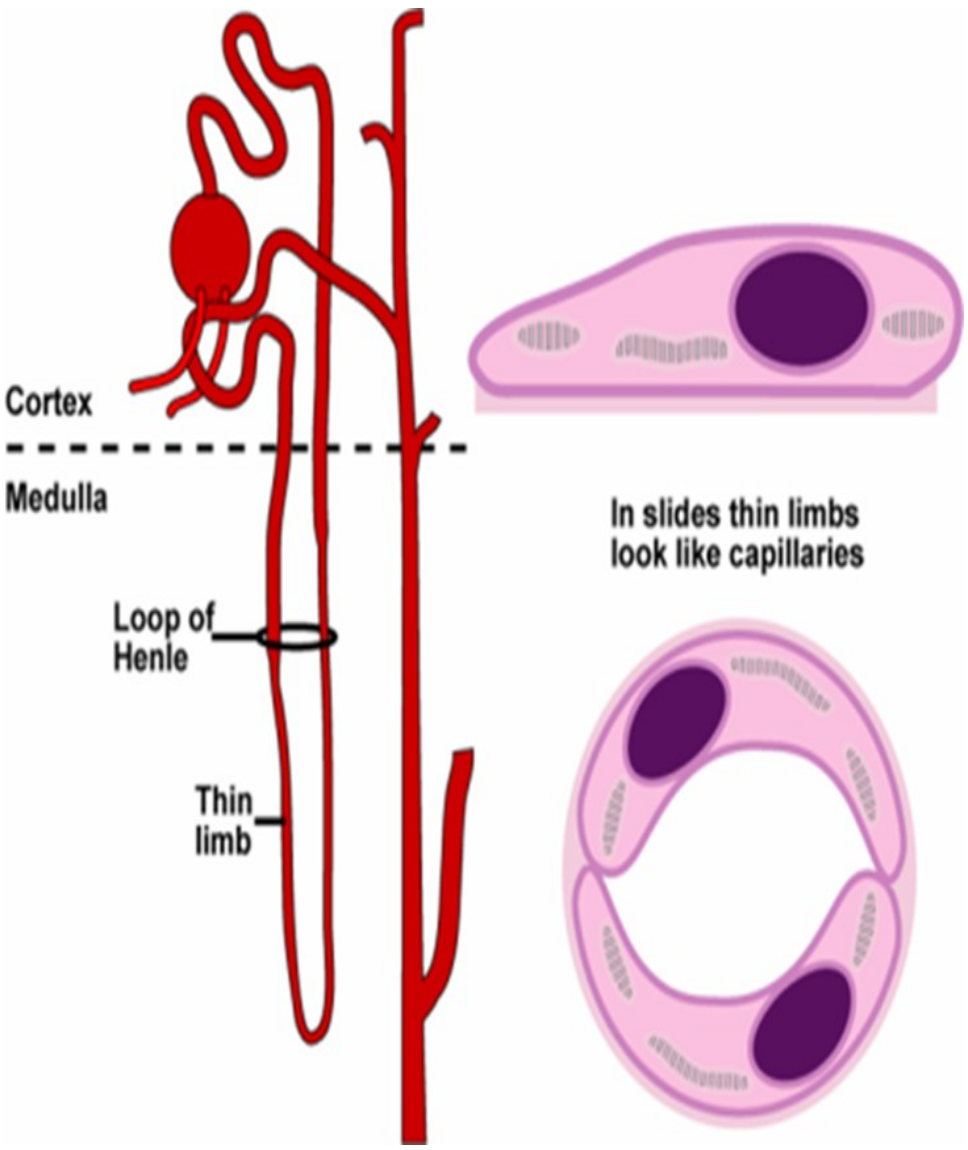
Lined by **simple squamous epithelium**

Function concentration of urine.



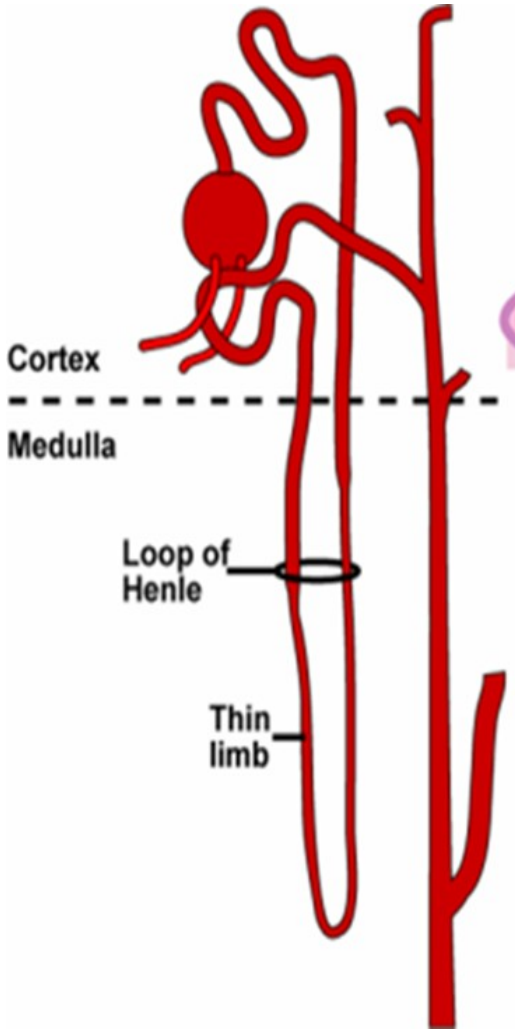
# Distal convoluted tubule

**Distal convoluted tubule** is highly tortuous about 5mm long and 25-50µ in diameter
Lined by simple cuboidal epithelium.
Basal striations are present indicating of ion transporting function of this tubule.
Reabsorb sodium chloride.
Sodium absorption occur under influence of aldosterones released by suprarenal cortex.



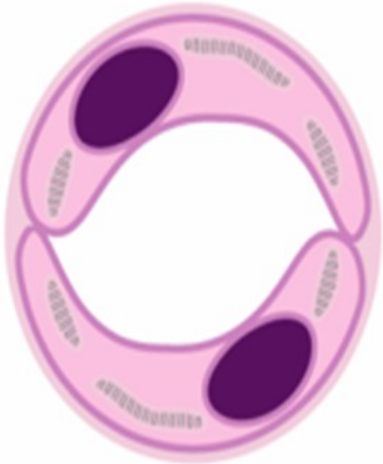
# Collecting Tubule

- Are excretory tubules, **not part of nephrons.**
- Collecting tubules is about 40 micrometer in diameter. Lined by **simple cuboidal epithelium**.
- **Collecting tubules open into Collecting Ducts**
- In medullary pyramid many collecting ducts join to form papillary ducts (of bellini) which opens into apex of papilla.






In slides thin limbs look like capillaries



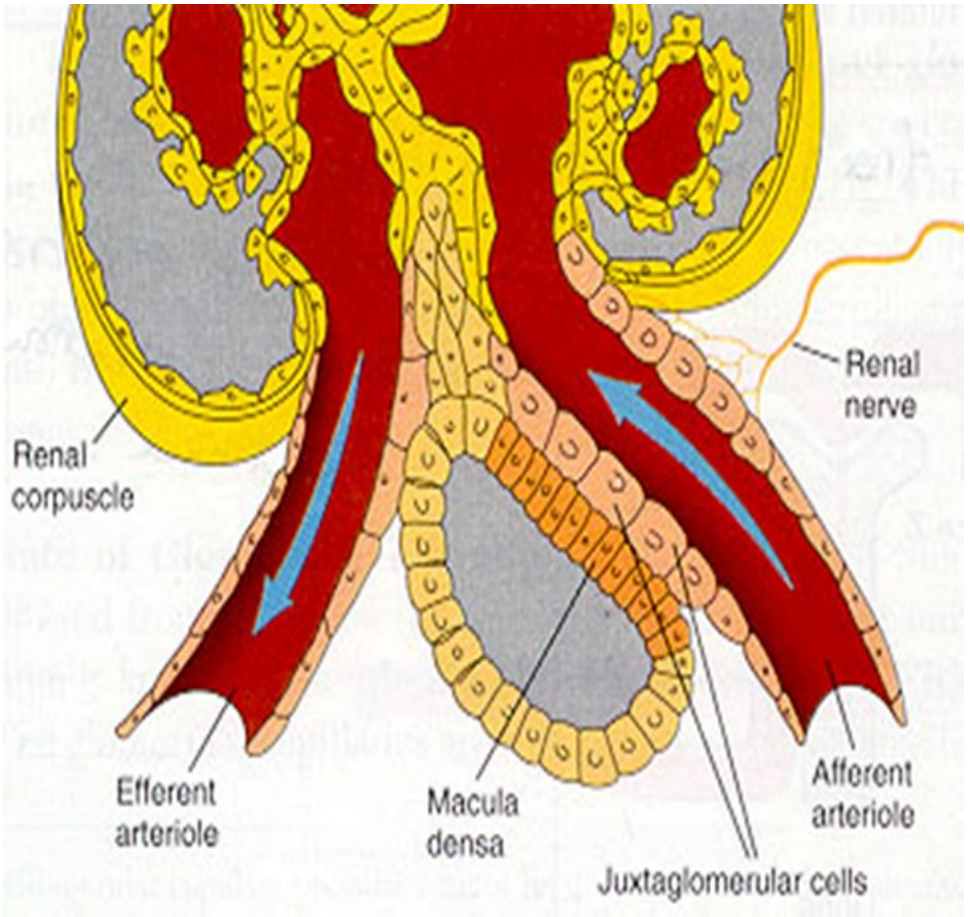
# Juxtaglomerular Complex

At the vascular pole of the renal corpuscle there is an association of three structure known as the juxtaglomerular complex .

1: Macula densa of the distal convoluted tubule and functions as a **sensor of osmolarity**.

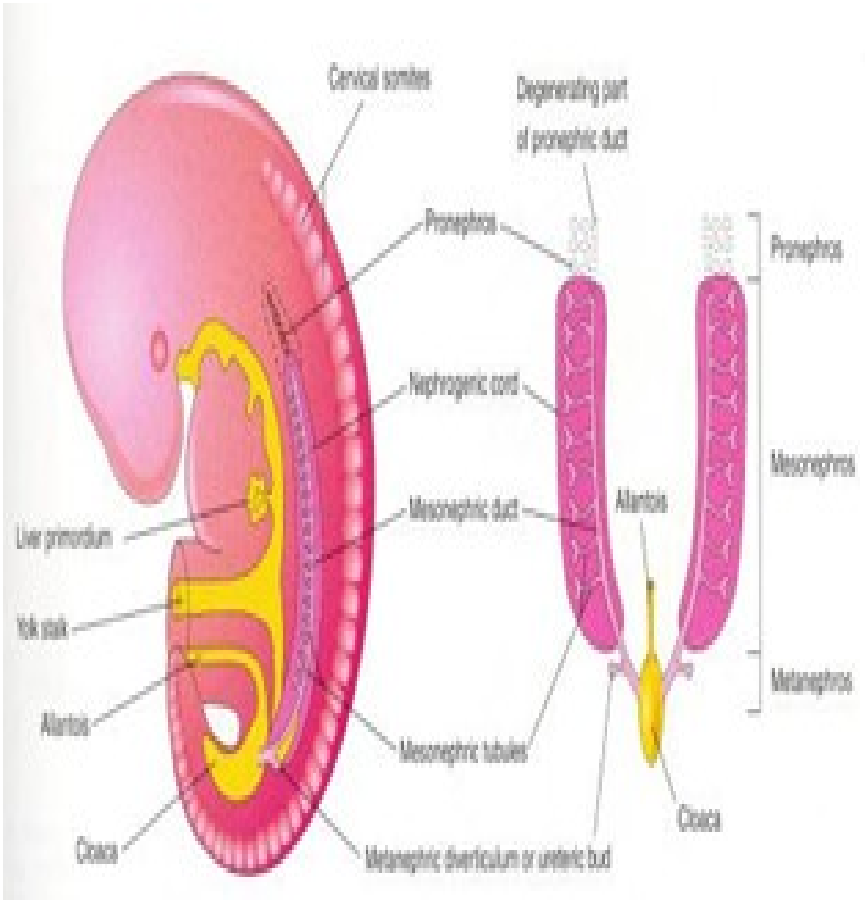
2:Juxtaglomerular cells in the wall of the afferent arteriole and produce a hormone called renin.

3:Extraglomerular mesengial cells situated between the afferent efferent arterioles of the glomerulus, provide structural support for capillary loops also are contractile and phagocytic .



# Development of Kidney

- Urinary system develops from intermediate mesoderm along with genital system.
- After 4th week the intermediate mesoderm forms a longitudinal mass on each side of primitive aorta in the trunk region called urogenital ridge.
- The part of urogenital ridge from which urinary system arise is known as nephrogenic cord and the part that gives rise to genital system is known as gonadal ridge.
- The excretory system of both urinary and genital system initially enter a common cavity called cloaca



Three sets of kidneys develop in human embryo

- Pronephros: Transient & developed in cervical region
- Mesonephros: Temporary and developed below the pronehros
- Metanephros: Permanent Kidneys developed in pelvic region

# Pronephros

The pronephros is 7-9 solid cell groups arising from mesodermal cells of the intermediate cell mass in cervical region.

These are transitory, nonfunctional structures and entirely disappear by the end of 4th week

Consists of tubules which open into longitudinal duct, Pronehric duct which drain in cloaca

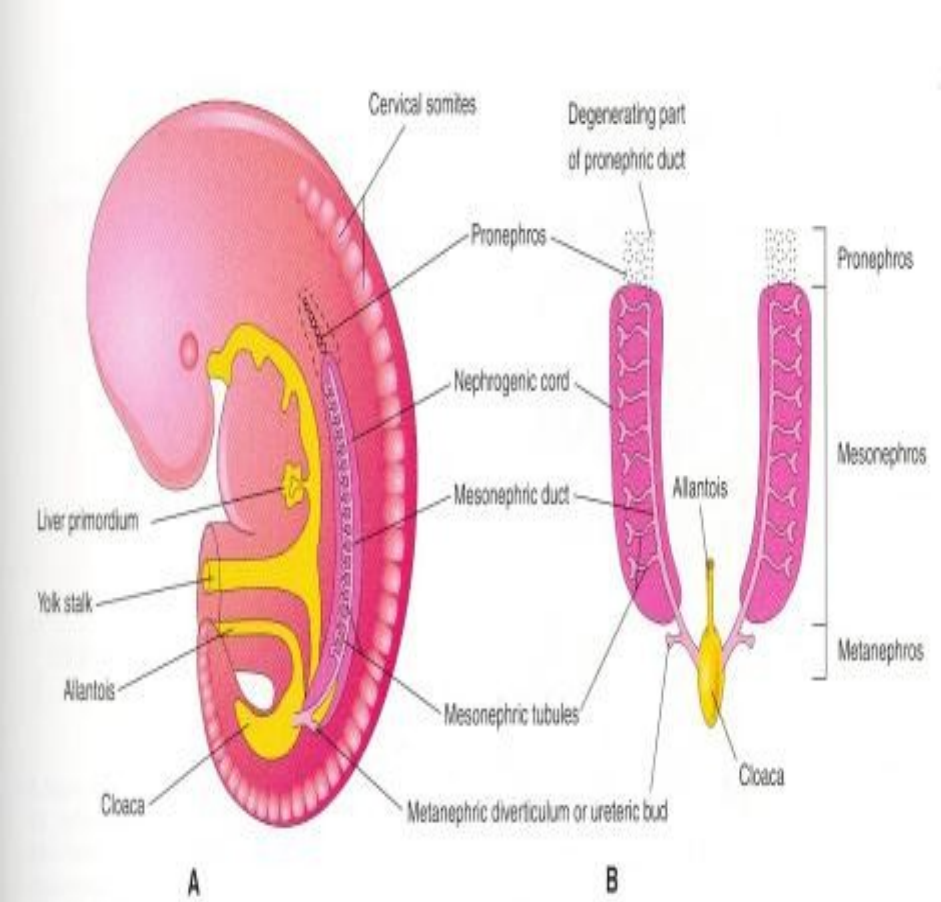



Figure 13–2. Diagrammatic sketches illustrating the three sets of excretory systems in an embryo during the fifth week. *A*, Lateral view. *B*, Ventral view. The mesonephric tubules have been pulled laterally; their normal position is shown in *A*.

# Mesonephros

The mesonephros is long elongated organ which appears in late 4th week caudal to rudimentary pronephros.

It is derived from intermediate cell mass from upper thoracic to upper lumber segments.

Mesonephric kidney consists of glomeruli and mesonephric tubules, which open into mesonephric duct (the pronephric duct becomes mesonephric duct .

Only few mesonephric tubules persists in male to participate in the formation of genital system.

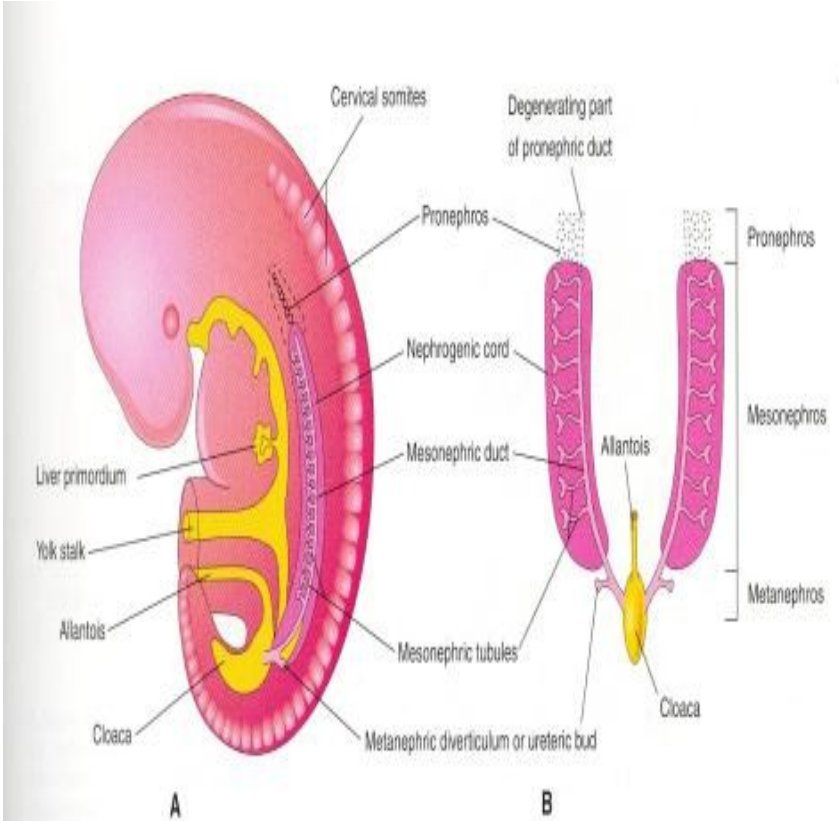



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# Metanephros

Metanephros the permanent kidney begins to develop early in the 5th week from metanepheric mesoderm.

It develops from two sources i.e. ureteric bud and metanephrogenic cap

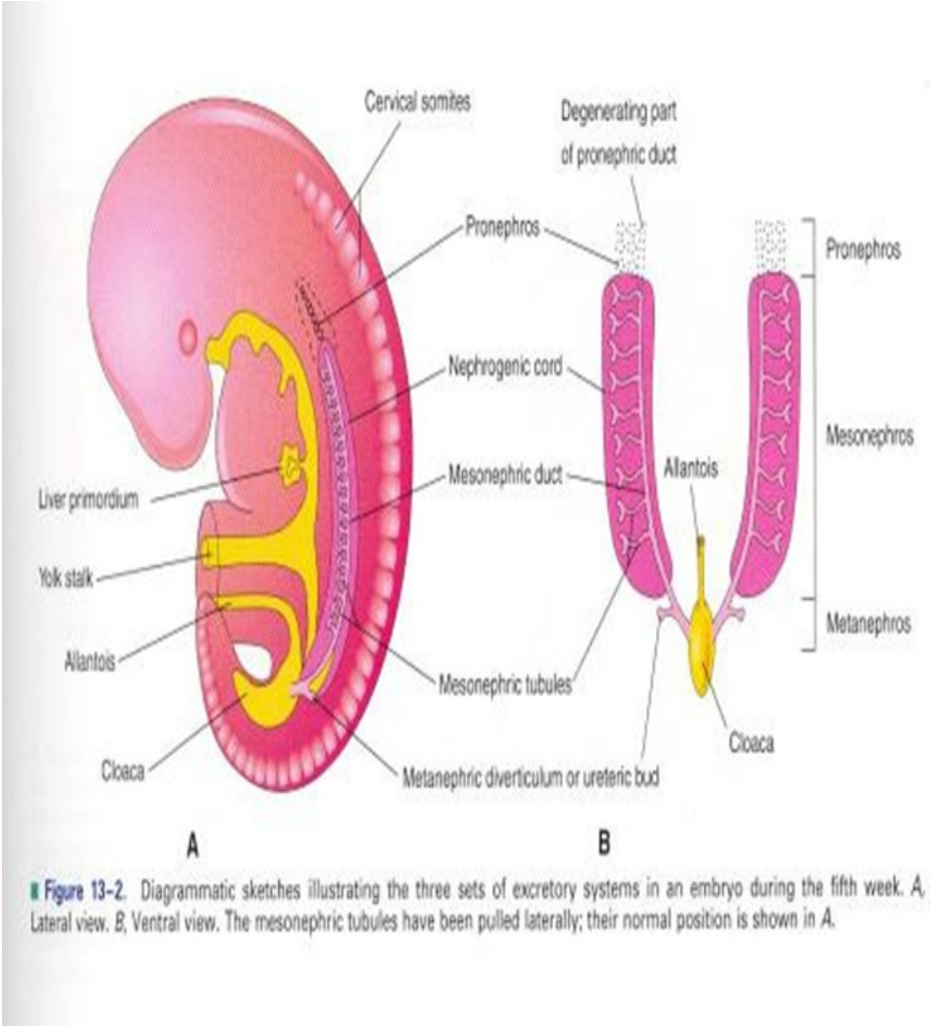



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# Metanephros

Ureteric bud forms first three to four generations of tubules which enlarge and become confluent to form major calyces and second four generation coalesce to form minor calyces and remaining generations of tubules form from collecting tubules of permanent kidney

Metanephrogenic cap is condensed mass of mesoderm with in the nephrogenic cord and form bowman's capsule proximal and distal convoluting tubules and loop of Henle.

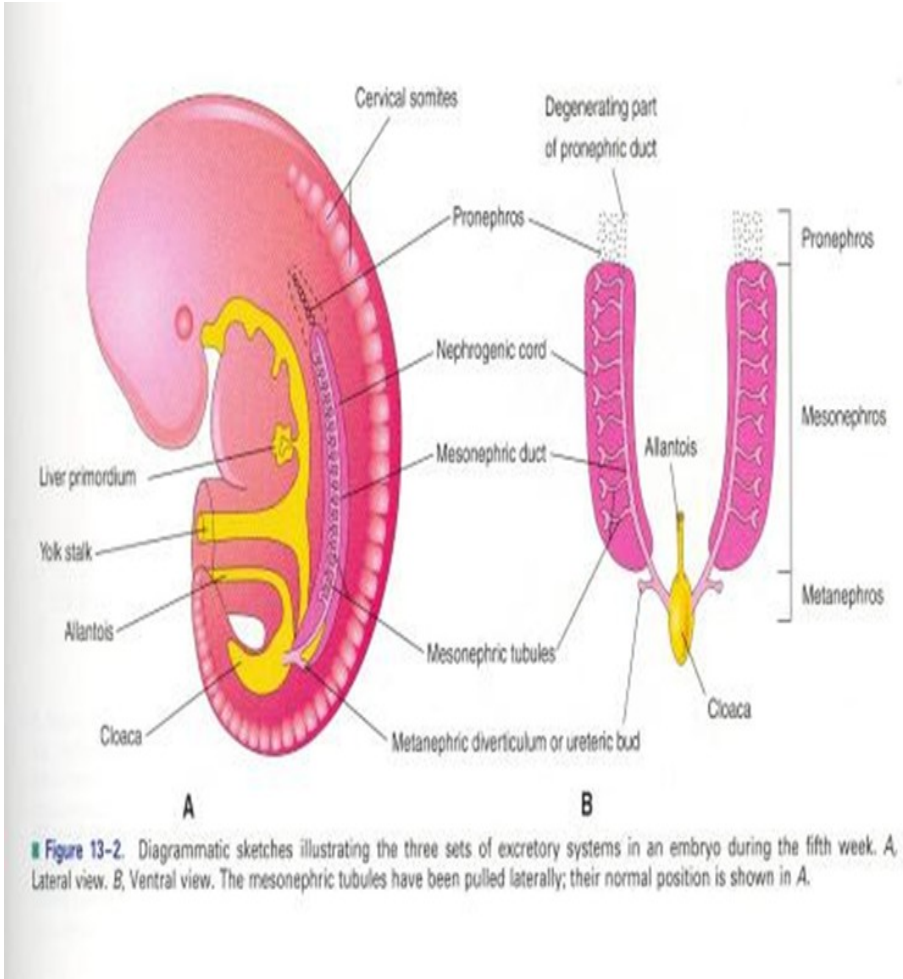



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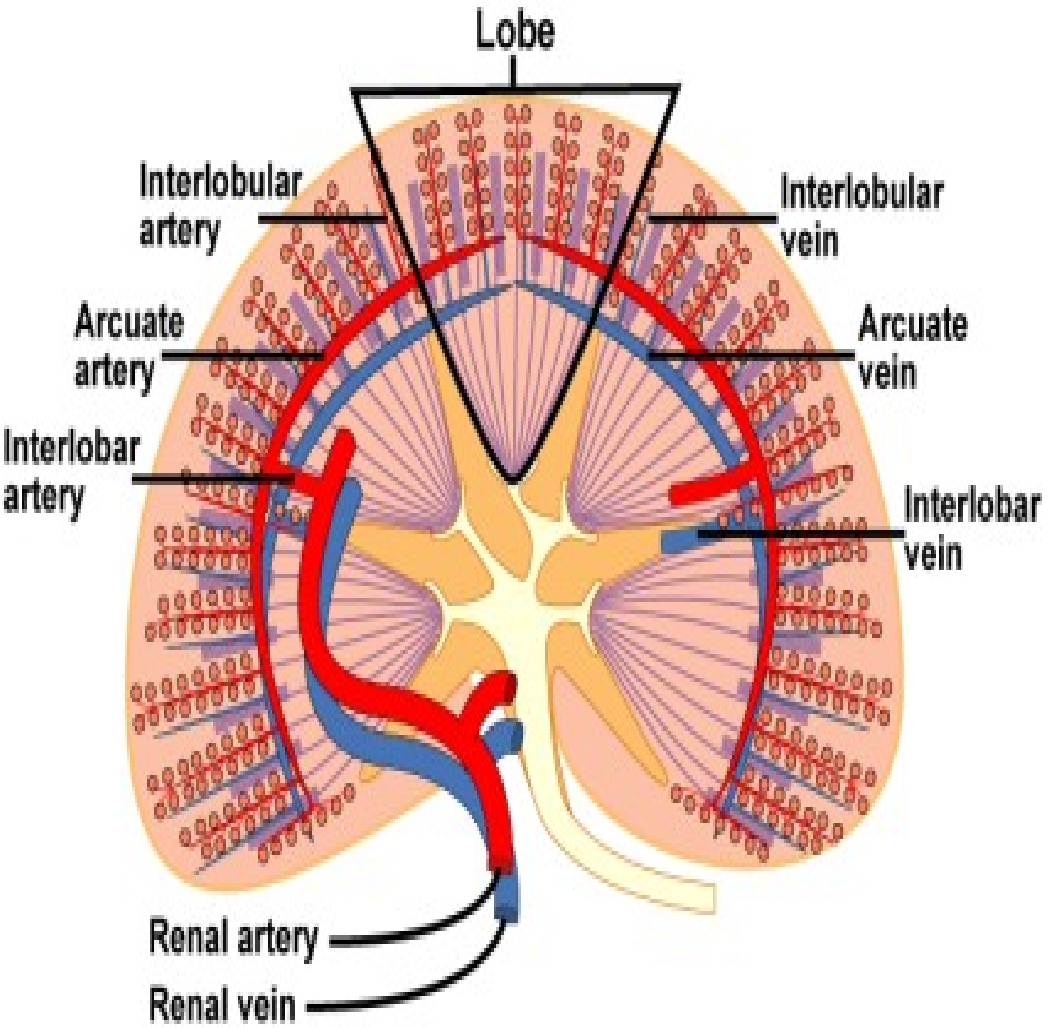
# Blood Supply of Kidney

Renal artery divides in to two or three main branches which divide into **interlobar arteries.**

At the corticomedullary junction each interlobar artery divides in to **arcuate arteries** that arch over the bases of the pyramids and run parallel to the surface of kidney.

From the **arcuate arteries** gives rise to interlobular arteries which travel through the cortex toward the renal corpuscle.

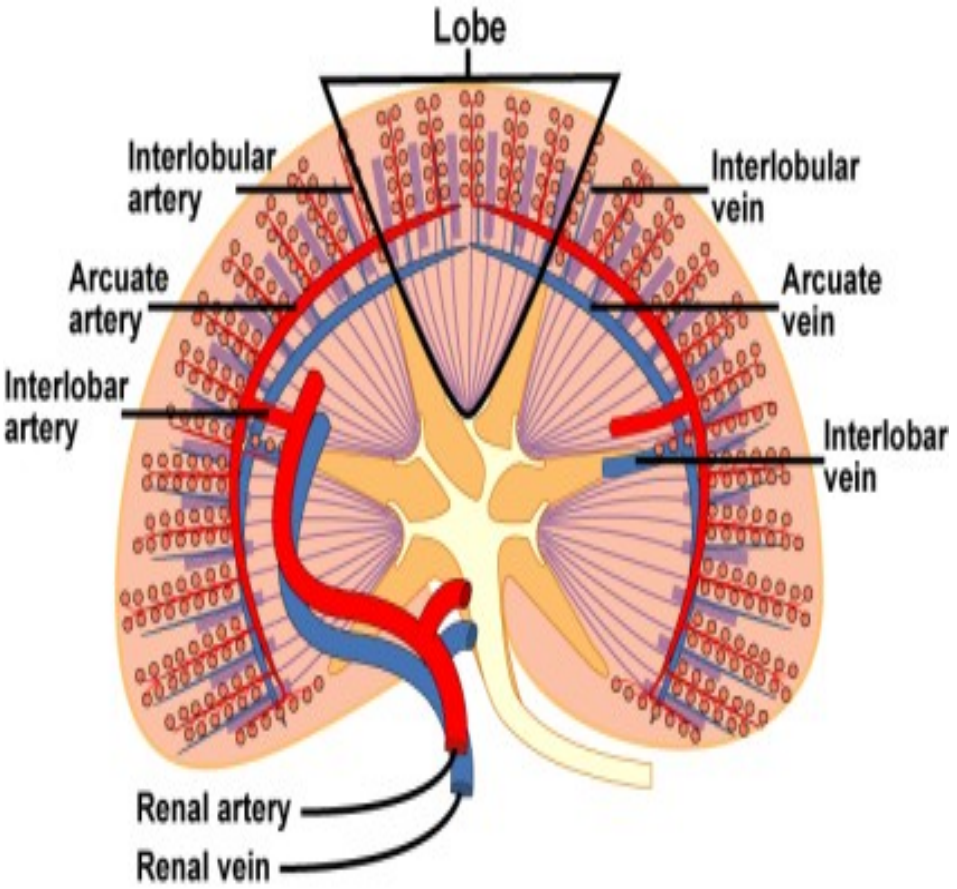
The interlobular arteries give rise to **intralobular branches** each of which become an afferent arteriole supplying the glomerular capillaries of a renal corpuscle.



# Blood Supply of Kidney

The glomerular capillaries re-unite to form the **efferent arteriole**.

Which branch again to form a peritubular capillary plexus that nourishes the proximal and distal convoluted tubules in the renal cortex and form long straight vessels  arteries are known as vasa recta.




## VENOUS DRAINAGE:

Vessel join to form **interlobular veins** which drain into **arcuate veins.**

Arcuate vein in turn open into **interlobar veins** which forms **renal vein**

**LYMPHATIC DRAINAGE:** Lateral aortic nodes located at the level of renal arteries

**NERVE SUPPLY:** Supplied by renal plexus which consists of sympathetic T10 - L1 fibers which are chiefly vasoconstrictor. The afferent fibers of kidney belong to segments T10-T12.

- **URETERS**: Superiorly continuous with the renal pelvis
- Inferiorly pass through the abdominal cavity, behind the peritoneum, in front of the psoas muscle, into the pelvic cavity
- Where they enter the posterior wall of the bladder
- 25-30 cm in length

**URINARY BLADDER**

Stores urine

- **URETHRA:** Transports urine from bladder to outside of body
- Difference in length between males and females.
- Female: 4cm long. Male: 14cm long
- Sphincters
- Internal urinary
- External urinary